

Defining the pollinator garden: is conceptual flexibility a feature or a bug?

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Ecologists often aim to reduce conceptual ambiguity by attempting to create rigid shared lexicons. These efforts imply that ambiguity is undesirable. In some contexts, however, conceptual flexibility comes with under-discussed benefits. Here, we use the lens of pollinator gardening to explore how conceptual flexibility is built into participatory conservation practices, and how this flexibility is central to the effectiveness of these practices. Grassroots conservation movements like pollinator gardening often have diffuse, bottom-up origins, producing a cloud of heterogeneous practices under an umbrella definition. We posit that conceptual flexibility provides key benefits in participatory conservation by attracting a larger, more diverse set of voluntary participants and experts, each of whom takes small actions that scale-up to large impacts in a process of conservation coproduction. We also suggest a flexible operational framework for ecologists studying participatory conservation, with the aim of enabling rigorous research without the need for rigid global definitions.

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Ecologists often seem to be engaged in a perpetual struggle against conceptual ambiguity. From addressing the “terminological tempest” in invasion science (Soto *et al.* 2024) to

tackling “widespread confusion” in community ecology (Stroud *et al.* 2015), the ecological literature is replete with calls for the standardization of definitions. In contrast, philosophers of language typically hold that such top-down standardization is impossible to achieve because language usage cannot be effectively controlled (Hodges 2008). Nonetheless, efforts at standardization among experts remain common because of the perceived costs of miscommunication and confusion. This article emerged from a working group in which we, a set of ecologists and practitioners, attempted to define the “pollinator garden”, an emerging topic of research in ecology and a popular conservation practice, with the aim of outlining goals and knowledge gaps for the field. Here, we describe how we came to realize that in this case a rigid definition, beyond being unachievable from the perspective of linguistics, would also have been counterproductive for conservation and research.

Pollinator gardening is a rapidly growing participatory conservation practice that exemplifies both the challenge and opportunity of conceptual flexibility. Participatory conservation involves mass participation by many people, each undertaking small conservation actions. In these movements, the specter of ambiguity seems to loom large because there is a diversity of perspectives and expertise in large groups, raising the likelihood of conflict in aims and interpretation of ambiguous terms. However, there are also underappreciated benefits to this kind of conceptual flexibility. In our deliberations over the definition of pollinator gardening, we recognized that, for participatory conservation, extensive conceptual flexibility not only is intrinsic but also promotes conservation efficacy and research progress. To illustrate our perspective, we first briefly review the origins of pollinator gardening, then identify the

In a nutshell:

- Though often seen as undesirable by scientists, flexible definitions are unavoidable; however, for participatory conservation, which relies on public participation, conceptual flexibility yields underappreciated benefits
- The concept of pollinator gardening exemplifies these benefits: rapidly spreading, inclusive, and adaptable, it accommodates a diverse set of practices under an umbrella term, which improves communication and yields broader appeal
- For participatory conservation movements like pollinator gardening, widespread appeal is vital to achieve the greatest impact
- We suggest ecologists recognize the benefits of flexibility for participatory conservation concepts, while precisely outlining subjects of research within these diverse movements

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Figure 1. Examples of the wide range of pollinator-friendly landscaping types that could be included (or not included) under the umbrella term “pollinator garden”. (a) An intentional pollinator garden, planted specifically to provide diverse nectar and pollen forage for pollinators. (b) A weedy yard, which either intentionally or unintentionally provides forage. (c) A residential prairie restoration, designed to mimic the ecological function of natural areas. (d) An ornamental flower garden, typically planted primarily for aesthetic reasons but also providing floral resources for pollinators. Image credits: (a) Matthew Shepherd, (b) Nicholas Dorian, (c) Heather Holm, and (d) Atticus Murphy.

possible costs of having an ambiguous definition, and later identify the advantages of conceptual flexibility from the perspectives of promoting conservation and of conducting research. We end by proposing a framework for ecologists to move forward with rigorous research in the face of conceptual flexibility.

■ Origins of pollinator gardening

The pollinator gardening concept broadly involves creating habitat for pollinators within urban and built environments (Figure 1). It emerged from the confluence of the gardening community, with its preexisting tradition of noticing and appreciating flower visitors (Habeck 1985; Curtin and Fox 2014), and the modern conservation movement, which has become increasingly concerned with insect declines and the loss of pollination services (Potts *et al.* 2010; Wagner 2020; Dorian *et al.* 2025). Interest in pollinator gardens is booming due to factors including popular campaigns like Save the [Honey] Bees (Hall and Martins 2020); mounting evidence of the potential conservation value of gardens for insects (Majewska and Altizer 2020); and benefits experienced by gardeners, including improved connection with nature and health outcomes (Raymond *et al.* 2019). To date, more than

one million pollinator gardens have been installed in the US (NPGN 2019).

The pollinator gardening concept did not originate in a centralized or formal way, but instead came about as a loose set of ideas that emerged organically. The bottom-up process leading to these practices is typical of the development of other kinds of participatory greening, such as volunteer tree plantings and community gardening (sometimes collectively termed civic ecology practices; Krasny *et al.* 2014, 2015). As volunteer-based, grass-roots movements, the implementation, scope, and goals of these practices are determined by the participants, who come from a diverse set of backgrounds and roles, including home gardeners, landscapers, and ecologists. Likely due to this process, the ideals and practices of pollinator gardening are heterogeneous and may be subject to change over time. This heterogeneity presents both challenges and opportunities for scientists and practitioners wishing to study and disseminate knowledge about the concept of pollinator gardens or any other participatory conservation practice.

■ Conceptual flexibility creates unavoidable costs for participatory conservation

The usual reason that ecologists and conservationists (including the authors of this article, in our initial attempts to define pollinator gardens) give for attempting to standardize terminology is that there are costs to ambiguous definitions. For conservation in particular, ambiguity could impair the ecological efficacy of practices (eg by confusing efforts to mitigate invasive species impacts; Humair *et al.* 2014; Mattingly *et al.* 2020). These costs might even be more substantial for participatory conservation because of the large number of people involved and the lack of centralized organization inherent to these grassroots movements.

One possible cost is that broad definitions imply broad (or even unspecified) goals. Vaguely specified goals create challenges for developing effective practices and could contribute to failure to recognize trade-offs between different goals. For pollinator gardens, there is little agreement on what the specific aims of pollinator gardening are, or how achievable different goals might be, which makes evaluating success a challenge (Dorian *et al.* 2025). Furthermore, gardening participants are likely to have varied priorities and motivations for their activities (Goddard *et al.* 2013; Philpott *et al.* 2020); some of these coexisting goals may align with conservation outcomes (eg colorful and diverse floral plantings), whereas others may not (eg desire for “neat” low-cut aesthetics)

(cf Dearborn and Kark [2010] on this issue in urban conservation broadly).

In addition, loosely defined concepts are at risk of semantic drift, in which the original meaning of a term is eventually lost, a concern previously raised about the terms “ecological restoration” (Woodworth 2017), “ecological integrity” (Rohwer and Marris 2021), and “extinction” (Smith-Patten *et al.* 2015). In conservation, this kind of gradual broadening in usage could eventually cause an expansion of on-the-ground activities considered part of the concept, potentially reducing the overall effectiveness of the conservation practice. For highly participatory conservation concepts like pollinator gardening, it is particularly difficult to influence which practices spread and become popular, because guidelines are not disseminated from a centralized source. As a result, excessive conceptual flexibility could open the door to the eventual inclusion of gardening practices with poor conservation value, in which volunteer effort is misspent on ineffective activities, such as the planting of cultivars with doubled flowers (characterized by extra petals and little pollen or nectar for pollinators) (Corbet *et al.* 2001), or the installation of dubiously useful butterfly “hibernation boxes”.

■ Conceptual flexibility also provides distinctive benefits to participatory conservation

Given these pitfalls, it might be natural to assume that pollinator gardening would be a more effective conservation practice if it had been more narrowly defined to meet ecological goals from the outset. This interpretation, however, neglects several important ways that participatory conservation concepts benefit from their inherent ambiguity and flexibility. We argue that definitional flexibility will (1) help participatory conservation movements grow and become more ecologically effective and (2) better enable researchers to study and make successful contributions to these coproduced participatory conservation concepts.

Conceptual flexibility increases ecological impact of participatory conservation

A flexible definition can increase uptake by being more inclusive of different practices and attracting a wider set of participants, each of whom brings their own visions, constraints, goals, and knowledge. For instance, a narrow definition for pollinator gardens might focus on native species, but a flexible definition could also include gardens lacking an emphasis on native species, which broadens appeal because sentiments about native plants vary widely among the general public (Rihn *et al.* 2024). The ability of pollinator gardening to attract large numbers of voluntary, novice participants is in many ways its greatest strength. Flexibility in practices is also likely to increase the equity of access to gardening, for example by reducing the initial cost of participation, which is likely a barrier to many potential gardeners (Knapp *et al.* 2020). In addition, because

meaningful habitat creation through gardening must involve many gardens rather than just one (Goddard *et al.* 2010; Belaire *et al.* 2014), attracting participants increases the value of all gardens. Although it is also useful for ecologists and conservation organizations to provide advice about how to make gardens more valuable or attractive to pollinators, these guidelines need not exclude gardens that have some but not all features that benefit pollinators. Accommodating some ecologically “suboptimal” idiosyncratic management decisions within individual gardens (like the inclusion of non-native flowering plants that are non-invasive) might instead be worthwhile, if it increases overall gardening uptake.

A flexible definition allows the practices of the pollinator gardening movement to evolve and adapt, incorporating regionally appealing elements and also adding or subtracting elements over time (cf Hodges 2008). For instance, over the past decade, the priorities of the pollinator gardening movement have shifted to include conserving the entire pollinator life cycle and not just the conspicuous adult life stages. As a result, advice is now widespread for gardeners to “leave the leaves” (let fallen leaves remain on the ground’s surface to promote insect development), create “soft landings” (diverse plantings at the base of a tree for overwintering insects; eg Holm 2025; Xerces Society 2025), or integrate artificial nesting structures such as “bee hotels” (MacIvor 2017). Because there is little top-down control over how gardeners interpret or implement these suggestions, participants are free to adopt new practices as they are developed.

Researchers benefit from explicitly recognizing the flexible nature of participatory conservation

A flexible definition helps researchers understand the true diversity of participatory conservation activities, as opposed to only a constrained subset thereof. In participatory conservation, thousands of gardeners become “land managers” from an ecological perspective; participatory conservation concepts are thus the result of a coproduction process (Beier *et al.* 2017) among these many gardeners and scientists. By embracing flexible definitions, rather than attempting to adopt rigid universal definitions, researchers can appropriately recognize the dynamics that underlie participatory conservation: a collaboration with a broad group of participants, and a movement that naturally shifts over time in composition and emphasis. This allows for research and guidance not only to adapt as new information emerges but also to learn from gardeners about their motivations, constraints, and insights in a dialogue, which is a more effective way to influence gardening practices (van Heezik *et al.* 2012). In addition, if researchers adopt operational definitions of an emerging concept (such as pollinator gardening) that are too rigid and narrow, we may constrain our ability to keep up with changing practices over time, and limit the kinds of questions that can be asked (Strunz 2012).

A flexible definition can promote communication among scientists as well as between scientists and participants. Even if

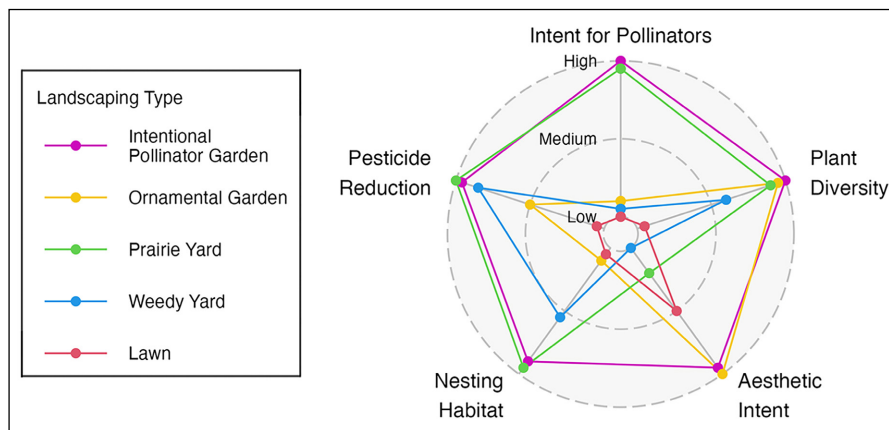


Figure 2. Conceptual illustration of certain dimensions that might be of interest to different researchers of pollinator gardens. In this radar plot, colors indicate yard landscaping types, whereas solid circles close to and far from the center have a low value and a high value, respectively, for a given dimension (values depicted here are for illustrative purposes only and simplify the true variability within yard types). Although, from the perspective of pollinator habitat, grass lawns are clearly inferior to the four other yard types (pictured in Figure 1), different researchers might include or exclude different yard types from their operational definition in researching pollinator gardens, depending on their focus. For example, ecologists studying habitat for pollinators might use an operational definition specifying low pesticide use, high plant diversity, and high-quality insect nesting habitat. Alternatively, from the perspective of research on the social practice of gardening, intent for creating habitat for pollinators and plans to construct an aesthetically appealing space might also be important elements of an operational definition, encompassing a different set of gardening practices.

we were to advocate for adopting a precise definition of an ideal pollinator garden (as defined by researchers), this definition may not wholly align with pollinator gardens as implemented by the gardening public (Dorian *et al.* 2025; see Heger *et al.* [2013] for discussion of similar problems in relation to invasive species). For example, while ecologists might identify key practices involving planting particular native plant species to support native bees in gardens, the perception of pollinators among gardeners in the US is instead largely dominated by the image of non-native, managed honeybees, whose habitat requirements are distinct from and more generalized than those of most other pollinators (Wilson *et al.* 2017; Hall and Martins 2020). Raising awareness of pollinator biodiversity and making these distinctions could be a suitable long-term conservation goal for pollinator gardens; however, in the meantime, pollinator gardening might benefit most from leveraging existing connections between people and garden insects that are widely recognized, rather than alienating potential participants. From the perspective of scientific progress, dealing with the true diversity of on-the-ground implementations of pollinator gardening helps researchers capture the broad and varied nature of these movements.

Defining pollinator gardens

Despite these benefits for conservation impact and research, conceptual flexibility still poses a challenge for researchers

of pollinator gardening and other flexible participatory concepts. In seeking to address this challenge, we recognized that a flexible, broad definition does not mean that research itself must be imprecise. In any given project, researchers should specify an operational definition of a pollinator garden: that is, an explicit definition of how pollinator gardens were defined in the context of a particular study (Heger *et al.* 2013). In this way, researchers can respond to the dynamic change of a grassroots movement while avoiding comparing apples to oranges (or sunflowers to petunias). This approach benefits research by acknowledging the true diversity of participatory conservation practices and helps to free ecologists from a (likely futile) struggle to more narrowly define the concept of pollinator gardening.

Our working group discussed a variety of possible definitions for pollinator gardens, comprising a sample of the many possible landscaping types and practices that could be studied by specialist researchers. We illustrated these operational definitions as encompassing variation along different conceptual axes depending on the research focus

(Figure 2). For example, one operational definition of a pollinator garden might focus on the dimensions of intention for aesthetic design, diversity of plant species, and intention to support pollinators. An “ideal” pollinator garden might then be a space that is designed and cultivated with pollinators in mind, emphasizing intentionality to support pollinators through plant selection and garden stewardship. Another operational definition might reduce the emphasis on intentionality for pollinators and aesthetic design and focus more on ecological impacts, broadening the focus of research to include weedy lots and remnant or restored habitat patches, and especially emphasizing pollinator-oriented management such as provision of overwintering habitat or reductions in pesticide use. Still another definition might primarily focus on garden aesthetic and flower diversity, thereby also including diverse ornamental gardens that provide floral resources but are not explicitly managed for pollinators.

In considering these examples, we do not intend to limit the dimensions of possible definitions. Instead, our goal is to embrace the flexibility and diversity of these practices while maintaining rigor (see Staples *et al.* [2025] for another framework-based approach to maintaining clarity and comparability across studies of novel ecosystems, a similarly heterogeneous concept). Importantly, to ensure that findings are comparable and that decisions about scope are appropriate, we advocate that researchers of pollinator gardening and other participatory practices explicitly report the criteria

(dimensions) used, as well as their rationale for the inclusion or exclusion of those criteria, in each of their studies. In this way, flexible overall concepts and situationally precise terminology can coexist to allow researchers to follow the unfolding of participatory conservation practices and be inclusive of their many forms, while precisely identifying the portions of these practices that are being considered in a specific study.

Conclusions

The ecologist's impulse to be suspicious of ambiguity is understandable. In the case of participatory conservation, however, conceptual flexibility yields both ambiguity and distinctive benefits. For pollinator gardening, this flexibility is one of the factors that has attracted such a wide range of both participants and researchers, thereby considerably increasing the potential ecological impact of this practice. For researchers, recognizing this flexibility is important to understand the heterogeneity of on-the-ground practices. In this way, pollinator gardens resemble many new and emerging fields of scientific inquiry, such as the "gene" concept, which has been flexible enough to continuously accommodate new understandings over decades (Hodges 2008). We suggest that because of their unusual process of development and implementation, participatory conservation practices are an especially important area in which to acknowledge the usefulness of flexibly defined concepts.

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No data were generated or analyzed.

References

- Beier P, Hansen LJ, Helbrecht L, and Behar D. 2017. A how-to guide for coproduction of actionable science. *Conserv Lett* 10: 288–96.
- Belaire JA, Whelan CJ, and Minor ES. 2014. Having our yards and sharing them too: the collective effects of yards on native bird species in an urban landscape. *Ecol Appl* 24: 2132–43.
- Corbet SA, Bee J, Dasmahapatra K, et al. 2001. Native or exotic? Double or single? Evaluating plants for pollinator-friendly gardens. *Ann Bot-London* 87: 219–32.
- Curtin S and Fox D. 2014. Human dimensions of wildlife gardening: its development, controversies and psychological benefits. In: Dixon GR and Aldous DE (Eds). *Horticulture: plants for people and places*, volume 3: social horticulture. Dordrecht, the Netherlands: Springer.
- Dearborn DC and Kark S. 2010. Motivations for conserving urban biodiversity. *Conserv Biol* 24: 432–40.
- Dorian NN, Murphy AW, Iler AM, and CaraDonna PJ. 2025. Setting goals for pollinator gardens. *Conserv Biol* 39: e70009.
- Goddard MA, Dougill AJ, and Benton TG. 2010. Scaling up from gardens: biodiversity conservation in urban environments. *Trends Ecol Evol* 25: 90–98.
- Goddard MA, Dougill AJ, and Benton TG. 2013. Why garden for wildlife? Social and ecological drivers, motivations and barriers for biodiversity management in residential landscapes. *Ecol Econ* 86: 258–73.
- Habeck DH. 1985. Attracting insects for backyard entomology. *Fla Entomol* 68: 117–21.
- Hall DM and Martins DJ. 2020. Human dimensions of insect pollinator conservation. *Curr Opin Insect Sci* 38: 107–14.
- Heger T, Saul W-C, and Trepl L. 2013. What biological invasions "are" is a matter of perspective. *J Nat Conserv* 21: 93–96.
- Hodges KE. 2008. Defining the problem: terminology and progress in ecology. *Front Ecol Environ* 6: 35–42.
- Holm H. 2025. Soft landings. www.pollinatorsnativeplants.com/softlandings.html. Viewed 6 Apr 2026.
- Humair F, Edwards PJ, Siegrist M, and Kueffer C. 2014. Understanding misunderstandings in invasion science: why experts don't agree on common concepts and risk assessments. *NeoBiota* 20: 1–30.
- Knapp JL, Phillips BB, Clements J, et al. 2020. Socio-psychological factors, beyond knowledge, predict people's engagement in pollinator conservation. *People Nat* 3: 204–20.
- Krasny ME, Russ A, Tidball KG, and Elmqvist T. 2014. Civic ecology practices: participatory approaches to generating and measuring ecosystem services in cities. *Ecosyst Serv* 7: 177–86.
- Krasny ME, Silva P, Barr C, et al. 2015. Civic ecology practices: insights from practice theory. *Ecol Soc* 20: 12.
- MacIvor JS. 2017. Cavity-nest boxes for solitary bees: a century of design and research. *Apidologie* 48: 311–27.
- Majewska AA and Altizer S. 2020. Planting gardens to support insect pollinators. *Conserv Biol* 34: 15–25.
- Mattingly KZ, Pelletier TA, Lanterman J, et al. 2020. Disconnects between communicated impact and ecological impact of biological invasions. *BioScience* 70: 252–63.
- NPGN (National Pollinator Garden Network). 2019. The Million Pollinator Garden Challenge meets its mark. <http://millionpollinorgardens.org/wp-content/uploads/2019/02/Million-Pollinator-Garden-Challenge-Report-FINAL-FOR-WEB-022519v3.pdf>. Viewed 6 Apr 2026.
- Philpott S, Egerer M, Bichier P, et al. 2020. Gardener demographics, experience, and motivations drive differences in plant species richness and composition in urban gardens. *Ecol Soc* 25: 8.
- Potts SG, Biesmeijer JC, Kremen C, et al. 2010. Global pollinator declines: trends, impacts and drivers. *Trends Ecol Evol* 25: 345–53.

- Raymond CM, Diduck AP, Buijs A, et al. 2019. Exploring the co-benefits (and costs) of home gardening for biodiversity conservation. *Local Environ* **24**: 258–73.
- Rihn AL, Torres A, Behe BK, and Barton S. 2024. Unwrapping the native plant black box: consumer perceptions and segments for target marketing strategies. *HortTechnology* **34**: 361–71.
- Rohwer Y and Marris E. 2021. Ecosystem integrity is neither real nor valuable. *Conserv Sci Pract* **3**: e411.
- Smith-Patten BD, Bridge ES, Crawford PHC, et al. 2015. Is extinction forever? *Public Underst Sci* **24**: 481–95.
- Soto I, Balzani P, Carneiro L, et al. 2024. Taming the terminological tempest in invasion science. *Biol Rev* **99**: 1357–90.
- Staples TL, Blois J, Cramer KL, et al. 2025. A conceptual framework for measuring ecological novelty. *Glob Ecol Biogeogr* **4**: e70005.
- Stroud JT, Bush MR, Ladd MC, et al. 2015. Is a community still a community? Reviewing definitions of key terms in community ecology. *Ecol Evol* **5**: 4757–65.
- Strunz S. 2012. Is conceptual vagueness an asset? Arguments from philosophy of science applied to the concept of resilience. *Ecol Econ* **76**: 112–18.
- van Heezik YM, Dickinson KJM, and Freeman C. 2012. Closing the gap: communicating to change gardening practices in support of native biodiversity in urban private gardens. *Ecol Soc* **17**: 34.
- Wagner DL. 2020. Insect declines in the Anthropocene. *Annu Rev Entomol* **65**: 457–80.
- Wilson JS, Forister ML, and Carril OM. 2017. Interest exceeds understanding in public support of bee conservation. *Front Ecol Environ* **15**: 460–66.
- Woodworth P. 2017. Can ecological restoration meet the twin challenges of global change and scaling up, without losing its unique promise and core values? *Ann MO Bot Gard* **102**: 266–81.
- Xerces Society. 2025. Leave the leaves: winter habitat protection. Portland, OR: The Xerces Society for Invertebrate Conservation. <https://xerces.org/leave-the-leaves>. Viewed 6 Apr 2026.

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